Clustering, Prototype Narratives, and Stiefel-Manifold Orthogonalization: A Worked Methodology

Applied to a Corpus of CNN Live-Blog Coverage of the 2026 Iran War

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following Anglade-Lehalle-Massi approach of orthogonalisation of narratives

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Abstract

This report documents the analytical pipeline applied to a corpus of 1,572 text snippets extracted from CNN's live-blog coverage of the US–Israel–Iran conflict (March–August 2026). The pipeline (i) embeds and clusters the snippets, in their *natural, unmodified embedding space*, into eight thematic groups; (ii) for each cluster, defines a small set of *representative narratives* (short illustrative sentences) and averages them into a single, more stable *prototype of narrative*; (iii) removes the mutual overlap between the eight prototypes by solving an optimization problem on the Stiefel manifold, yielding what we call *pure narratives* — versions of each prototype purged of the "background noise" shared by all clusters (ordinary English grammar and the generic vocabulary of Iran-war reporting); and (iv) scores every snippet against the raw and pure prototypes to measure how well each representation discriminates between themes. Because the pure narratives no longer carry that shared background component, variation in a document's cosine similarity to a pure narrative over time reflects variation in its *relative exposure to that specific narrative*, rather than drift in generic war-reporting language. The central empirical result is that the pure (orthogonalized) prototypes separate in-cluster from out-of-cluster documents markedly better than the raw prototypes, which serves as an *ex post*, label-free validation of both the clustering and the orthogonalization procedure. All steps are illustrated in detail on a single cluster, *Maritime Conflict & Strait of Hormuz*.

1 Overview of the pipeline

The analysis proceeds in five stages, detailed in the sections that follow:

Stage 1. Embed and cluster the corpus in its natural space. Every paragraph-level snippet is embedded with a sentence-transformer and grouped into $K = 8$ clusters using K-Means directly on this embedding space (after a variance-preserving PCA reduction, purely for numerical stability). *No orthogonalization or other transformation is applied at this stage*: the clustering reflects the raw geometry of the embedding model.

Stage 2. Define representative narratives and average them into a prototype. For each cluster, the analyst curates a small number of short, illustrative sentences (here, three, generated with the help of Gemini) called *representative narratives*. Because a single representative narrative can be idiosyncratically worded, the three are embedded and *averaged* (then re-normalized) into one more stable vector per cluster, the *prototype of narrative*.

Stage 3. Orthogonalize the prototypes into "pure" narratives. The eight prototype embeddings are projected onto an orthonormal basis via a custom `stiefel_optimize` routine. The result is called a *pure narrative*: a version of the prototype from which the "background noise" common to *all* clusters — generic English sentence structure and the shared vocabulary of Iran-war reporting — has been removed, leaving (approximately) only what is specific to that one theme.

Stage 4. Score every document under three representations. Each of the 1,572 snippets is compared (cosine similarity) against the *original* prototypes, the *pure (orthogonal)* prototypes, and the *residual* (original − pure) component — the part of each prototype made up of exactly the shared background noise removed in Stage 3.

Stage 5. Validate ex post. The separation between in-cluster and out-of-cluster similarity is compared across the three representations. Sharper separation under the pure-narrative representation is taken as evidence that the clustering captures real, distinguishable narrative axes, and that changes in similarity to a pure narrative over time reflect genuine shifts in a document's exposure to that narrative rather than shared background drift.

2 Stage 1: Embedding and clustering the corpus

The raw corpus consists of daily CNN live-blog pages, each split into paragraph-level snippets after removing boilerplate (navigation text, "Get our five biggest...", duplicated live-coverage banners, etc.), leaving $n = 1{,}572$ snippets spanning March 1 – August 18, 2026.

2.1 Embedding

Each snippet s_i is mapped to a unit-norm vector $\mathbf{e}_i \in \mathbb{R}^{768}$ using a sentence-embedding model (`nomic-ai/modernbert-embed-base`), so that cosine similarity between two snippets reduces to a simple dot product:

$$\mathrm{sim}(s_i, s_j) = \frac{\mathbf{e}_i^\top \mathbf{e}_j}{\|\mathbf{e}_i\|\,\|\mathbf{e}_j\|} = \mathbf{e}_i^\top \mathbf{e}_j \quad (\text{since } \|\mathbf{e}_i\| = 1).$$

2.2 Dimensionality reduction and clustering

Important methodological point: clustering is performed entirely in the *natural embedding space* of the snippets, i.e. on the raw sentence-transformer output (after the linear, variance-preserving PCA step described below). No orthogonalization, background-noise removal, or any other transformation from later stages is applied before or during clustering. This matters for the validity of Stage 5: whatever structure K-Means finds is a property of the embedding model and the corpus alone, not an artifact of the orthogonalization machinery introduced afterwards.

Because K-Means and Euclidean distances behave poorly in very high-dimensional, near-uniform embedding spaces, the 768-dimensional embeddings are first projected to 50 dimensions with PCA. K-Means with $K = 8$ clusters is then fit on the PCA scores (*not* on the 2D t-SNE coordinates, which are used only for visualization and would distort distances). Each resulting cluster is inspected manually and given a short thematic label. Table 1 lists the eight clusters and their size; full descriptions, rationale, and illustrative excerpts for every cluster are given in Appendix A. The cluster used as a worked example throughout the main text is *Maritime Conflict & Strait of Hormuz*

($n = 215$), which groups reporting on shipping disruption, tanker attacks, naval blockades, and transit through the Strait of Hormuz.

Table 1: The eight K-Means clusters, their thematic label, and the number of text snippets assigned to each ($n = 1{,}572$ total). Full descriptions are given in Appendix A.

Cluster	Theme	Size
0	Gulf Regional Military Escalation	209
1	Global Energy Markets & Economic Fallout	81
2	Iranian Political Transition & Internal Affairs	110
3	US-Israel Military Campaign in Iran	246
4	US Domestic Politics & Presidential Leadership	306
5	Israel-Hezbollah-Lebanon Theater	170
6	Diplomacy & International Mediation	235
7	Maritime Conflict & Strait of Hormuz	215
	Total	**1,572**

3 Stage 2: Representative narratives and the prototype of narrative

3.1 Representative narratives

For each of the $K = 8$ clusters, three short sentences are curated as *representative narratives*: text generated with the help of Gemini and selected by the analyst to be illustrative of that theme. We denote the three representative-narrative embeddings for cluster i by $\mathbf{r}_{i,1}, \mathbf{r}_{i,2}, \mathbf{r}_{i,3} \in \mathbb{R}^{768}$. This choice is deliberately not automated (e.g., not the medoid or centroid of the cluster): it encodes the analyst's semantic understanding of what the theme *is*, independently of how K-Means happened to partition the data.

Worked example — representative narratives for *Maritime Conflict & Strait of Hormuz*

$\mathbf{r}_{7,1}$: Maritime traffic through the Strait of Hormuz has slowed to a near halt as shipping companies reroute or suspend passage.

$\mathbf{r}_{7,2}$: US forces intercepted and seized a cargo vessel attempting to bypass the naval blockade of Iranian ports.

$\mathbf{r}_{7,3}$: An oil tanker was struck and disabled by an unidentified projectile near the strait's entrance.

3.2 From representative narratives to a prototype of narrative

A single representative narrative can be idiosyncratically worded and is a noisy estimate of "what the theme is about" in embedding space. To obtain a more stable anchor, the three representative-narrative embeddings of each cluster are *averaged and re-normalized* into one vector, which we call

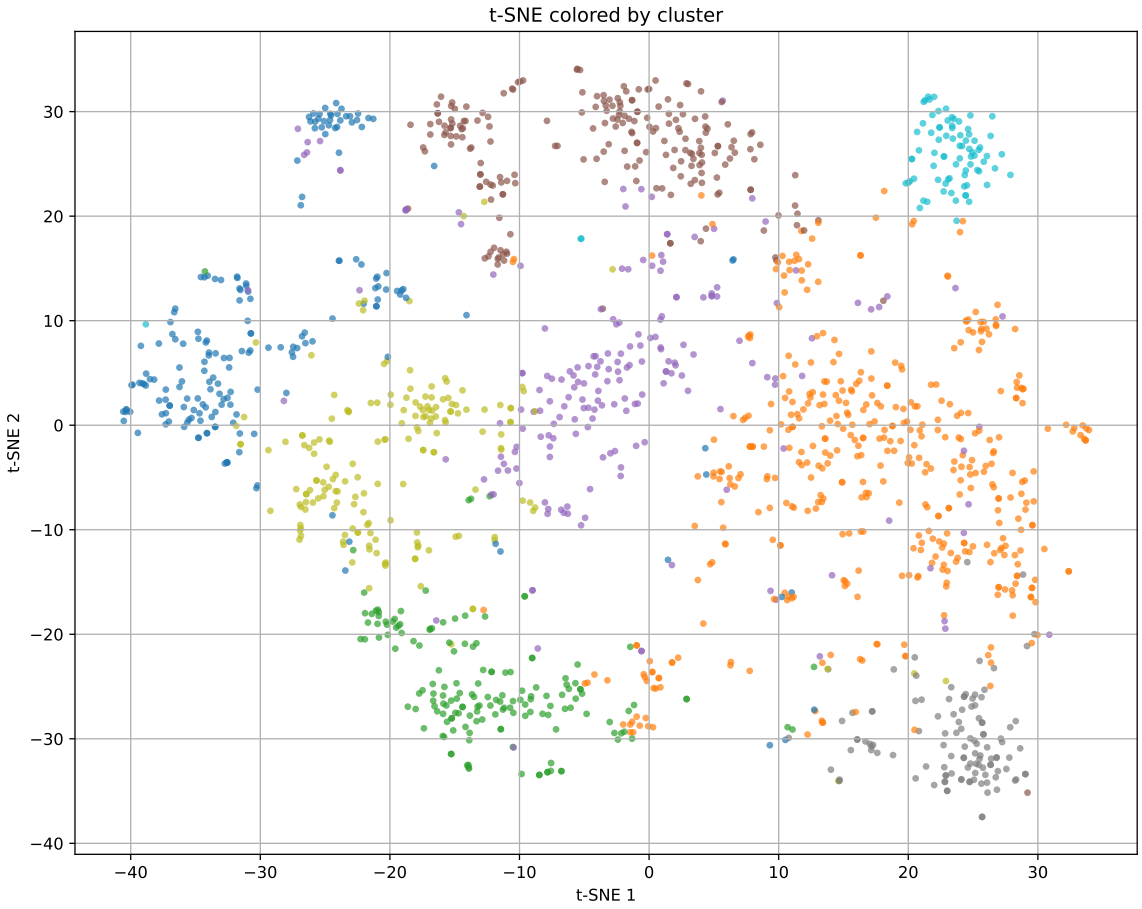


Figure 1: t-SNE projection (2 dimensions, computed on the same PCA-reduced, natural embedding space used for K-Means, cosine metric, perplexity ≈ 35) of all 1,572 snippets, colored by their K-Means cluster label. **This plot is diagnostic only**: t-SNE is used purely to *visualize* the clustering after the fact; K-Means itself, as stressed above, is fit on the 50-dimensional PCA scores, never on these 2D coordinates. Two qualitative features are worth noting. First, most clusters form spatially coherent, largely non-overlapping blobs — direct visual confirmation that the eight themes correspond to genuinely separable regions of the natural embedding space, consistent with the quantitative separation reported in Section 6. Second, the larger clusters (e.g. *US Domestic Politics*, $n = 306$, or *US-Israel Military Campaign*, $n = 246$) tend to appear as elongated or multi-lobed regions rather than single tight blobs, indicating that they are internally more heterogeneous, likely covering several recurring sub-topics under one broad label; smaller, more topically narrow clusters (e.g. *Global Energy Markets*, $n = 81$) instead form a single dense clump. Scattered points sitting between two blobs are boundary snippets that share vocabulary with more than one theme — exactly the kind of shared background noise that the Stiefel orthogonalization (Section 4) is designed to factor out when scoring documents against a narrative.

the **prototype of narrative** for that cluster:

$$\mathbf{N}_i \;=\; \frac{\frac{1}{3}\sum_{k=1}^{3}\mathbf{r}_{i,k}}{\left\|\frac{1}{3}\sum_{k=1}^{3}\mathbf{r}_{i,k}\right\|}, \qquad i = 1, \ldots, 8,$$

giving an 8×768 matrix $\mathbf{N}$ of prototypes across all clusters. It is this averaged prototype $\mathbf{N}_i$ — not the individual representative narratives $\mathbf{r}_{i,k}$ — that is used as the input to the orthogonalization step (Stage 3) and to all downstream scoring (Stage 4). Averaging three phrasings of the same idea cancels out some of each individual sentence's idiosyncratic wording while keeping the direction that is common to all three, which is precisely the theme itself.

4 Stage 3: Orthogonalization on the Stiefel manifold

4.1 Motivation: separating signal from shared background noise

Figure 2 (top-left 8×8 block) shows the pairwise cosine similarity between the eight prototypes of narrative $\mathbf{N}_i$ built in Stage 2. Off-diagonal values of 0.6–0.8 are common: prototypes for genuinely distinct themes still point in very similar directions. We interpret this shared component as a **common background noise** made of two ingredients that every English sentence about this conflict carries, almost regardless of its specific theme: (a) ordinary *English grammar and sentence structure* (subject-verb-object patterns, connective words, tense markers), and (b) the *generic vocabulary of Iran-war reporting* shared across all eight themes (place names, military and diplomatic jargon, recurring proper nouns such as "Iran", "Trump", "strikes"). Because this background noise is present in *every* prototype, it is exactly the part of $\mathbf{N}_i$ that is problematic for discrimination: a snippet can score highly against several prototypes simultaneously merely because the prototypes themselves share this background noise, not because the snippet is thematically ambiguous. The goal of orthogonalization is to strip this shared component out and isolate what we call the **pure narrative** of each theme.

4.2 The optimization problem

The goal is to find, for each prototype $\mathbf{N}_i$, a new vector $\mathbf{V}_i$ such that:

- the set $\{\mathbf{V}_i\}$ is *exactly orthonormal*: $\mathbf{V}\mathbf{V}^\top = I$, i.e. $\mathbf{V}$ lies on the **Stiefel manifold** $\mathrm{St}(K, d) = \{V \in \mathbb{R}^{K\times d} : VV^\top = I_K\}$;
- each $\mathbf{V}_i$ remains *as close as possible* to its original direction $\mathbf{N}_i$, so that the orthogonalized narrative is still recognizably about the same theme.

Formally, writing $\langle \mathbf{N}_i, \mathbf{V}_i \rangle$ for the alignment (cosine similarity) between the i-th original and orthogonalized prototype, the routine `stiefel_optimize` solves

$$\max_{V \in \mathrm{St}(K,d)} \; \min_{i=1,\ldots,K} \langle \mathbf{N}_i, \mathbf{V}_i \rangle,$$

i.e. it maximizes the *worst-case* alignment across all K prototypes, so that no single narrative is sacrificed to improve the others. Because the hard min is non-smooth, the implementation uses a smooth *soft-min* surrogate with temperature τ (here $\tau = 0.05$):

$$\mathrm{softmin}_\tau(x_1, \ldots, x_K) = -\tau \log \sum_{i=1}^{K} \exp\!\left(-\frac{x_i}{\tau}\right), \qquad x_i = \langle \mathbf{N}_i, \mathbf{V}_i \rangle,$$

which is minimized (cost = −softmin) using Riemannian conjugate gradient descent (`pymanopt`'s `ConjugateGradient` optimizer), initialized from a Gram–Schmidt orthogonalization of $\mathbf{N}$. Convergence is fast: 24–40 iterations, a few seconds of wall-clock time.

4.3 Interpretation of the three vectors

For every cluster i, the procedure yields three related, comparable objects:

- **Prototype (original) $\mathbf{N}_i$**: the averaged, re-normalized representative-narrative embedding from Stage 2 — contains both theme-specific signal *and* the shared background noise (grammar + generic war vocabulary).
- **Pure narrative (orthogonal) $\mathbf{V}_i$**: the Stiefel-projected version, mutually orthogonal to all other $\mathbf{V}_j$ ($j \neq i$) by construction — our approximation of the narrative's content *net of the shared background noise*, i.e. what is genuinely unique to theme i.
- **Residual (background noise) $\mathbf{R}_i = \dfrac{\mathbf{N}_i - \mathbf{V}_i}{\|\mathbf{N}_i - \mathbf{V}_i\|}$**: the (renormalized) component removed from the prototype to obtain the pure narrative — our estimate of the shared grammar/vocabulary signal that theme i's prototype has in common with the other themes.

Why this matters for interpreting cosine similarity. Because the pure narrative $\mathbf{V}_i$ is (by construction) purged of the background noise common to all eight themes, a document's cosine similarity to $\mathbf{V}_i$ is not inflated or deflated by how much generic "war-reporting English" that document happens to contain. Consequently, when we later track $\mathbf{e}_t^\top \mathbf{V}_i$ over time or across documents (Stage 4–5), movements in that quantity can be attributed to genuine changes in the document's *relative exposure to the pure narrative i itself*, rather than to incidental drift in how much generic conflict vocabulary a given day's reporting happens to use. This is the key property that the original, non-orthogonalized prototypes $\mathbf{N}_i$ do *not* have: two documents could differ in $\mathbf{e}_t^\top \mathbf{N}_i$ simply because one is written in a more "typical war-report" style, with no difference in actual exposure to theme i.

Worked example — from prototype to pure narrative for *Maritime Conflict*

The prototype $\mathbf{N}_{\text{Maritime}}$ has cosine similarity ≈ 0.78 with $\mathbf{N}_{\text{US-Israel Military Campaign}}$ and ≈ 0.71 with $\mathbf{N}_{\text{Diplomacy}}$ (see Fig. 2, row *Maritime Conflict & Strait of Hormuz*, top-left block): both share heavy "military operations" and "negotiation" vocabulary — exactly the kind of background noise described above. After optimization, the pure narrative $\mathbf{V}_{\text{Maritime}}$ has, by construction, cosine similarity ≈ 0.00 with every other pure narrative $\mathbf{V}_j$ (Fig. 2, "orth" block, off-diagonal entries), while retaining an alignment of 0.77 with its own prototype direction $\mathbf{N}_{\text{Maritime}}$ (diagonal of the original×orth block): the pure narrative is still recognizably "about" maritime conflict, just stripped of what it shared with the rest of the corpus. The corresponding residual (background-noise estimate) $\mathbf{R}_{\text{Maritime}}$ instead correlates strongly *positively* with the other clusters' prototypes (0.51–0.67, bottom-left block of Fig. 2) and negatively with its own pure narrative (−0.48), confirming that it isolates exactly the shared, non-discriminative component that orthogonalization removed.

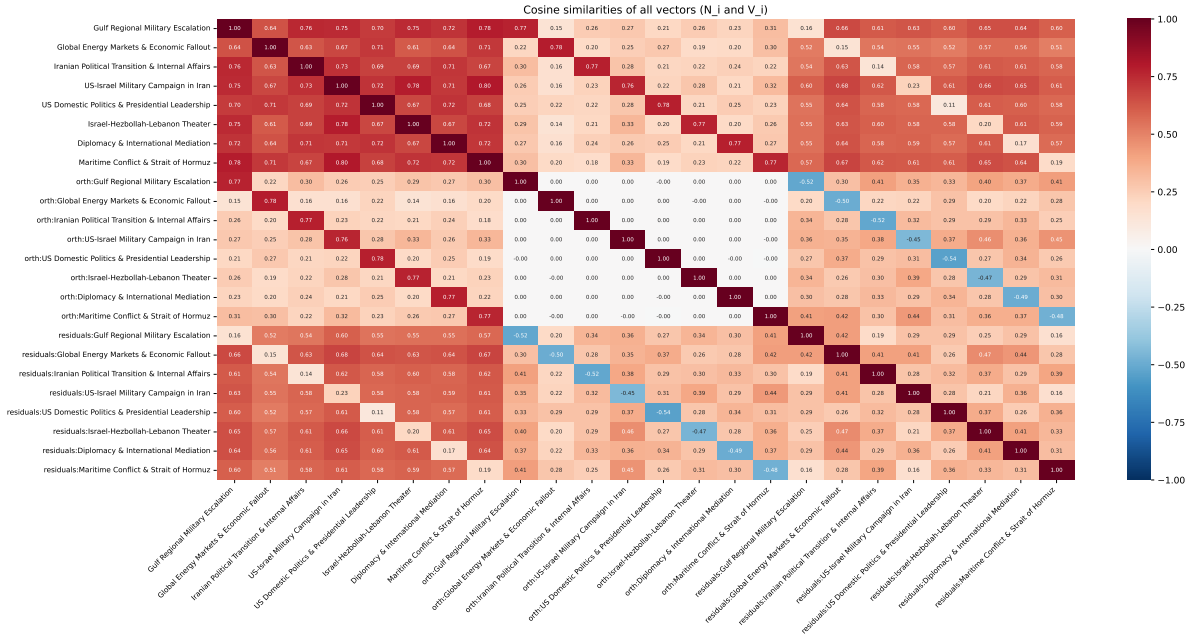


Figure 2: Pairwise cosine similarity between the 8 original prototypes (top-left block), the 8 orthogonalized prototypes (middle block, exactly block-diagonal by construction), and the 8 residual vectors (bottom-right block). Note how the original prototypes are highly inter-correlated (0.6–0.8), whereas the orthogonal ones are, by construction, uncorrelated with one another (off-diagonal ≈ 0) yet remain well aligned with their own original direction (values on the "original vs. orth" diagonal, ≈ 0.7–0.8).

5 Stage 4: Scoring the corpus under three representations

Every snippet s_t ($t = 1, \ldots, 1572$) already carries a K-Means cluster label $c(t) \in \{1, \ldots, 8\}$ from Stage 1 — a label obtained in the *natural embedding space*, before any orthogonalization existed. Independently of that label, its embedding $\mathbf{e}_t$ is scored against *all eight* prototypes under each of the three representations (prototype / pure narrative / residual), giving three 1572×8 similarity matrices:

$$S^{\text{orig}}_{t,i} = \mathbf{e}_t^\top \mathbf{N}_i, \qquad S^{\text{orth}}_{t,i} = \mathbf{e}_t^\top \mathbf{V}_i, \qquad S^{\text{res}}_{t,i} = \mathbf{e}_t^\top \mathbf{R}_i.$$

Recall that $S^{\text{orth}}_{t,i}$ is, by the argument of Section 4.3, a measure of document t's *relative exposure to the pure narrative of theme* i, cleaned of the shared background noise that would otherwise contaminate $S^{\text{orig}}_{t,i}$. For each snippet, two quantities are then extracted for each representation:

- **in-cluster similarity**: $S_{t,\,c(t)}$ — similarity to the prototype of the snippet's own K-Means cluster;
- **out-of-cluster similarity**: $\{S_{t,j} : j \neq c(t)\}$ — similarity to the seven other prototypes.

Worked example — one snippet from *Maritime Conflict*

Consider the snippet (assigned by K-Means to *Maritime Conflict & Strait of Hormuz*) reporting that only a handful of ships crossed the strait on a given day because captains and owners remained cautious. Its similarity profile is qualitatively:

Prototype (j)	S^{orig}	S^{orth}	S^{res}
Maritime Conflict (own cluster, $j = c(t)$)	~ 0.80	~ 0.52	~ 0.30
US-Israel Military Campaign	~ 0.75	~ 0.28	~ 0.55
Diplomacy & Mediation	~ 0.70	~ 0.22	~ 0.50

Under the *original* representation, the snippet's own theme wins, but only by a narrow margin over unrelated themes (0.80 vs. 0.70–0.75): the raw embedding space is "crowded". Under the *orthogonal* representation, the margin widens sharply in relative terms (0.52 vs. 0.22–0.28): once the shared conflict-reporting signal is projected out, the snippet is unambiguously closest to its own, purified theme vector.

6 Stage 5: Ex post validation

If the K-Means clustering (Stage 1) and the prototype selection (Stage 2) both captured a genuine underlying structure, we would expect the *orthogonalized* prototypes — which were built completely independently of the K-Means labels, from only 3 hand-picked sentences per cluster — to discriminate the 1,572 labelled documents *better* than the raw, correlated prototypes. This is exactly what is tested in Figures 4 and 5.

Worked example — reading the result for *Maritime Conflict*

From Figure 4 (column "Maritime Conflict & Strait of Hormuz"): under the *original* representation, in-cluster similarity (left column, top row) sits around 0.78–0.80 with essentially no overlap with the out-of-cluster distribution (right column, top row, centered ≈ 0.65) — already a reasonably clean separation, consistent with this being one of the most lexi-

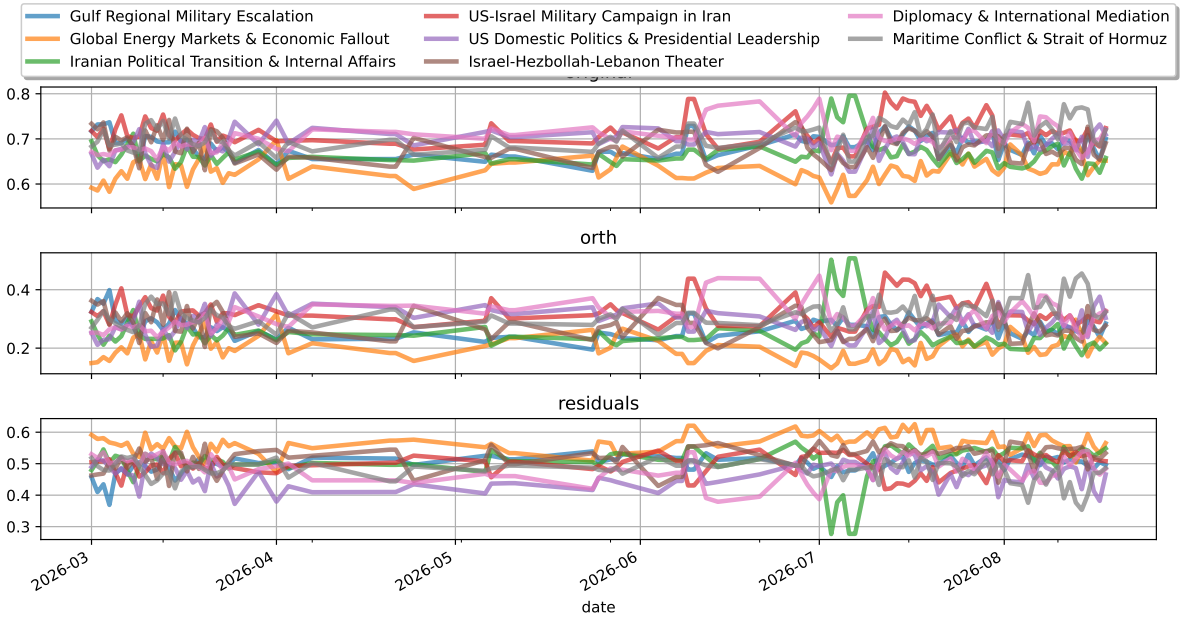


Figure 3: Daily average cosine similarity trajectories of the eight thematic narratives across the CNN corpus under the three representations: *original* prototypes (top), *orthogonal / pure* prototypes (middle), and *residuals* (bottom). Whereas the original and residual series co-move in a tight, highly correlated band, the orthogonalized representation isolates distinct, theme-specific spikes over time.

cally distinctive themes in the corpus (shipping, tankers, blockades). Under the *orthogonal* representation (middle row), in-cluster similarity drops in absolute terms (≈ 0.52) but the out-of-cluster distribution drops *much further* (≈ 0.27), so that the standardized gap (Fig. 5, "Maritime Conflict" row) is the *largest of all eight themes* (≈ 4.3 standard deviations, versus ≈ 3.5 for the original representation). This is consistent with *Maritime Conflict* being a theme with a genuinely distinct technical vocabulary ("Strait of Hormuz", "tanker", "blockade") that survives the orthogonalization essentially intact, while its correlation with other themes (through shared war-reporting language) is precisely what gets stripped out.

6.1 Take-away

Across all eight themes (Fig. 5), the pure-narrative (orthogonal) representation matches or exceeds the prototype's discriminative power, while the residual (background-noise) representation is uniformly non-discriminative (negative separation). Since the pure narratives were derived from only three representative sentences per cluster and an unsupervised geometric constraint (orthonormality), *not* from the K-Means labels themselves — and since K-Means itself was run on the untouched, natural embedding space (Section 2) — this constitutes an unsupervised, label-free corroboration that:

1. the eight K-Means clusters correspond to genuinely separable narrative axes in the natural embedding space, and

2. the Stiefel-manifold orthogonalization is not merely an algebraic transformation but recovers

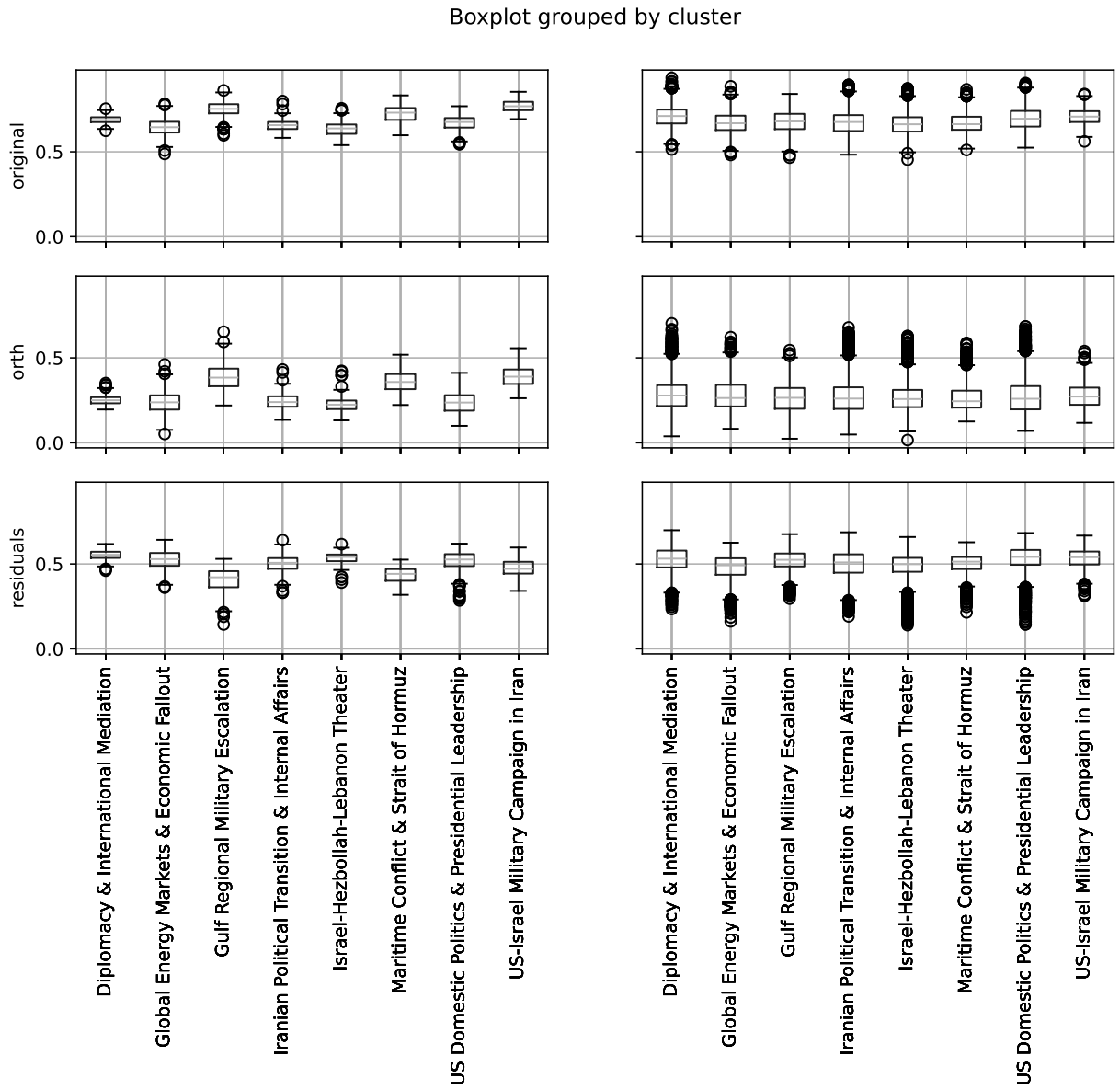


Figure 4: Distribution (boxplots) of cosine similarity scores, grouped by theme, under the three representations (rows: original / orthogonal / residual) and two conditions (columns: *in-cluster* similarity — a document scored against the prototype of its own K-Means cluster; and *out-of-cluster* similarity — the same document scored against the other seven prototypes). Compare the vertical gap between the left and right columns across rows: it is small for "original" and "residuals", and comparatively much larger for "orth".

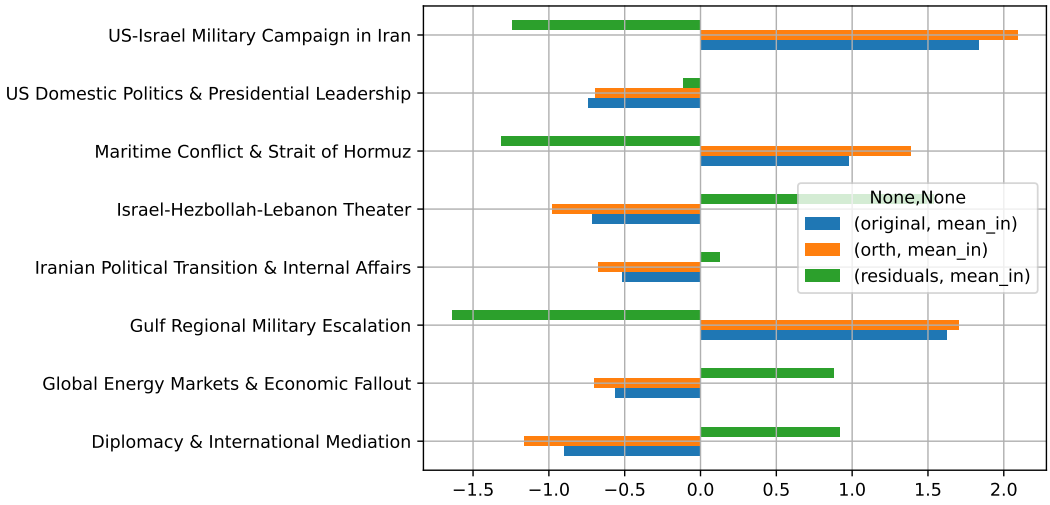


Figure 5: Standardized in-cluster separation, $(\overline{S}_{\text{in}} - \overline{S}_{\text{out}})/\sigma_{\text{in}}$, for each theme and each representation. Blue and orange bars (original vs. orthogonal) are of comparable magnitude and consistently *positive* for every theme, confirming discriminative power in both representations; the orange (orthogonal) bars are slightly larger than the blue (original) ones for every single theme, indicating a systematic — if moderate — improvement in separation once the shared component is projected out. Green bars (residuals) are uniformly *negative*: a document's similarity to the *residual* of its own cluster is, on average, *lower* than to the residuals of other clusters, which is the expected signature of a component that carries generic, non-discriminative signal rather than theme-specific signal.

additional, meaningful discriminative signal — the pure narrative — that was masked by shared background noise in the raw prototypes, and

3. consequently, tracking a document's (or a day's) similarity to a *pure* narrative over time is a meaningful way to monitor exposure to that specific narrative, largely free of confounding drift in generic war-reporting language.

Appendix

A Detailed cluster descriptions

This appendix gives, for each of the eight K-Means clusters (natural embedding space, Section 2), its thematic label, the rationale for that label, and a short illustrative excerpt of the kind of reporting it contains (paraphrased and abbreviated from the underlying CNN live-blog text; original snippet indices are given in brackets for traceability). Cluster sizes are repeated from Table 1 for convenience.

Cluster 0 — Gulf Regional Military Escalation ($n = 209$)
Theme: direct attacks, drone/missile interceptions, evacuations, and security alerts across secondary Gulf states (Saudi Arabia, Kuwait, UAE, Bahrain, Qatar, Iraq, Yemen). **Rationale:** distinct from the Iran-centered strike cluster (Cluster 3), this group captures how hostilities spread physically across the wider Gulf and Red Sea region, with an emphasis on civilian infrastructure, air-defense activations, and evacuations of foreign nationals. **Illustrative excerpts:** Saudi air defenses intercepting drones near Riyadh `[2026-03-02 #10]`; Qatar evacuating residents near the US Embassy as a precaution `[2026-03-04 #16]`; stranded travelers waiting out flight cancellations in Dhaka `[2026-03-01 #13]`.

Cluster 1 — Global Energy Markets & Economic Fallout ($n = 81$)
Theme: oil-price surges, gas-price spikes, market volatility, and government responses such as Strategic Petroleum Reserve releases. **Rationale:** isolates financial/economic consequences of the war (supply-chain and price effects) as distinct from the military actions that cause them. **Illustrative excerpts:** Trump downplaying oil prices above $100/barrel `[2026-03-08 #18]`; the SPR release of 172 million barrels `[2026-03-11 #15]`; a small manufacturer describing supply disruptions `[2026-03-21 #10]`.

Cluster 2 — Iranian Political Transition & Internal Affairs ($n = 110$)
Theme: Iran's internal governance, the death of Supreme Leader Ali Khamenei, the succession of Mojtaba Khamenei, and domestic repression. **Rationale:** groups reporting on internal regime dynamics that are largely orthogonal, substantively, to the external military campaign. **Illustrative excerpts:** the Assembly of Experts selecting a new supreme leader `[2026-03-08 #13]`; mass mourning crowds during the multi-day funeral `[2026-07-06 #5]`; reports of a wave of political executions `[2026-05-06 #4]`.

Cluster 3 — US-Israel Military Campaign in Iran ($n = 246$)
Theme: direct kinetic operations — airstrikes, missile exchanges, naval engagements — between US/Israeli forces and Iranian military targets, including CENTCOM operational updates. **Rationale:** this is the "main campaign" cluster, defined by strike counts, named targets (e.g. Kharg Island), and official military communiqués. **Illustrative excerpts:** over 3,000 targets struck under Operation "Epic Fury" `[2026-03-06 #8]`; strikes on Kharg Island oil infrastructure `[2026-03-13 #4]`; CENTCOM footage of naval strikes near Hormuz `[2026-03-10 #16]`.

Cluster 4 — US Domestic Politics & Presidential Leadership ($n = 306$)
Theme: White House rhetoric, presidential statements, press briefings, and congressional oversight debates. **Rationale:** this cluster is anchored in Washington politics — how the US executive communicates and defends its war strategy domestically — rather than in events on the ground. **Illustrative excerpts:** Trump outlining war objectives to reporters `[2026-03-02 #0]`; a White

House press briefing on administration resolve [2026-03-25 #13]; Senate Democrats forcing a war-powers vote [2026-03-09 #12].

Cluster 5 — Israel-Hezbollah-Lebanon Theater ($n = 170$)
Theme: the parallel Israel–Hezbollah/Lebanon (and adjacent Gaza) front: rocket and airstrike exchanges, ground movements, and fragile ceasefires. **Rationale:** separates the Mediterranean flank of the conflict from the Iran-centered campaign (Cluster 3), since the actors, geography, and ceasefire dynamics differ substantially. **Illustrative excerpts:** Israel intensifying strikes on Lebanon [2026-03-01 #7]; Hezbollah rocket fire damaging homes in central Israel [2026-03-12 #18]; a fragile Lebanon-Israel ceasefire under strain [2026-06-04 #10].

Cluster 6 — Diplomacy & International Mediation ($n = 235$)
Theme: diplomatic backchannels, mediated negotiations (chiefly by Qatar, Pakistan, and Oman), and the drafting of ceasefire or peace frameworks. **Rationale:** isolates official diplomatic statements and mediation efforts, as distinct from the military and domestic-political narratives. **Illustrative excerpts:** Pakistan offering to host US-Iran talks [2026-03-23 #12]; envoys Witkoff and Kushner meeting Qatar's prime minister [2026-06-30 #1]; a trilateral defense pact among Saudi Arabia, Turkey, and Pakistan [2026-08-07 #1].

Cluster 7 — Maritime Conflict & Strait of Hormuz ($n = 215$)
Theme: commercial shipping disruption, naval blockades, tanker attacks, and transit management in the Strait of Hormuz and the wider Persian Gulf. **Rationale:** a domain-specific cluster with a distinctive technical vocabulary (tankers, chokepoints, transit risk), used as the worked example throughout the main text. **Illustrative excerpts:** traffic through the Strait crawling to a near halt [2026-03-03 #18]; a container vessel damaged off the UAE coast [2026-03-10 #2]; US Marines seizing an Iranian-flagged cargo ship attempting to run the blockade [2026-04-20 #5].